

RESEARCH PREPRINT

The Dutch Disease Trap

An Empirical Evaluation of Export Complexity, Market Concentration, and
Commodity Lock-in in Kazakhstan

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Introduction

In recent decades, the economic development of Kazakhstan has been heavily reliant on its vast natural resource endowments, particularly crude oil, metals, and minerals. While resource abundance can accelerate initial economic growth, literature on the “**Resource Curse**” and “**Dutch Disease**” suggests that over-reliance on primary commodities can hinder the development of non-resource sectors.

According to modern trade theory pioneered by the Harvard Growth Lab and MIT, a country’s long-term economic prosperity depends not just on the volume of its exports, but on their **quality** and **complexity**.

Research Hypothesis

This study aims to evaluate the structural dynamics of Kazakhstan’s international trade. We formulate our core research hypothesis as follows:

Hypothesis (H_1): High dependence on primary commodity exports concentrates Kazakhstan’s trade profile and constrains the growth of its export economic complexity, making the economy vulnerable to global commodity price shocks.

To test this hypothesis, we will systematically analyze: 1. **Export Concentration:** Using the Herfindahl-Hirschman Index (HHI). 2. **Technological Intensity:** Assessing the share of high-tech exports over time. 3. **Economic Complexity:** Evaluating Kazakhstan’s position in the global Economic Complexity Index (ECI).

Data & Methodology

To ensure the reproducibility of this research, all data is fetched dynamically via open APIs from the World Bank indicators database.

Methodological Framework & Mathematical Formalization

To quantitatively evaluate Kazakhstan’s trade profile and test Hypothesis (H_1), we operationalize three foundational metrics from international trade theory and economic complexity literature:

1. Herfindahl-Hirschman Market Concentration Index (HHI)

Export product concentration is measured via the normalized Herfindahl-Hirschman Index:

$$HHI_t = \frac{\sum_{i=1}^N \left(\frac{X_{it}}{X_t}\right)^2 - \frac{1}{N}}{1 - \frac{1}{N}}$$

Where X_{it} denotes the export value of product category i at time t , X_t is total national aggregate export volume, and N represents the total number of traded product groups. $HHI \rightarrow 1$ signals extreme commodity-driven concentration, whereas $HHI \rightarrow 0$ reflects structural diversification.

2. Revealed Comparative Advantage (RCA) Framework

Following Balassa (1965), structural trade specialization across processing stages is identified using the Revealed Comparative Advantage index:

$$RCA_{it} = \frac{X_{i,KAZ,t} / X_{Total,KAZ,t}}{X_{i,World,t} / X_{Total,World,t}}$$

An $RCA_{it} > 1$ confirms an explicit comparative advantage in product group i . We evaluate the evolution of RCA across three processing tiers: raw agricultural inputs, mid-stream ores and minerals, and fully processed manufactured commodities.

3. Economic Complexity & Diversity Metrics

In accordance with Hidalgo and Hausmann (2009), the Economic Complexity Index (ECI) measures the knowledge intensity embedded within a nation's export matrix by solving the linear system connecting Diversity ($k_{p,0}$) and Ubiquity ($k_{c,0}$):

$$\widetilde{M}_{c,c'} = \sum_p \frac{M_{cp} M_{c'p}}{k_{c,0} k_{p,0}}$$

Where $M_{cp} = 1$ if $RCA_{cp} \geq 1$, and 0 otherwise. ECI corresponds to the eigenvector associated with the second largest eigenvalue of matrix $\widetilde{M}_{c,c'}$.

Environment Setup

First, we load the core libraries required for data manipulation, API querying, and visualization.

Data Acquisition

We extract two critical indicators from the World Bank World Development Indicators (WDI) database to capture the trade structure: 1. **Fuel Exports (% of merchandise exports)**: Captures the resource dependence (TX.VAL.FUEL.ZS.UN). 2. **High-technology Exports (% of manufactured exports)**: Represents the technological complexity and diversification alternative (TX.VAL.TECH.MF.ZS).

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Table 1: Kazakhstan Trade Structure Raw Data Subset (World Bank)

Year	Fuel Exports (% of Merchandise)	High-Tech Exports (% of Manufactured)
2010	71.68	35.36
2011	72.02	25.68
2012	69.88	31.01
2013	76.30	38.54
2014	76.39	39.26
2015	67.72	43.43
2016	60.73	32.10
2017	63.33	24.21
2018	70.03	23.57
2019	66.24	29.29

Visualizing Trade Structure Trends

To better understand the structural shifts in Kazakhstan's economy, we visualize the trajectories of resource dependence (fuel exports) versus high-tech manufacturing survival over the analyzed period.

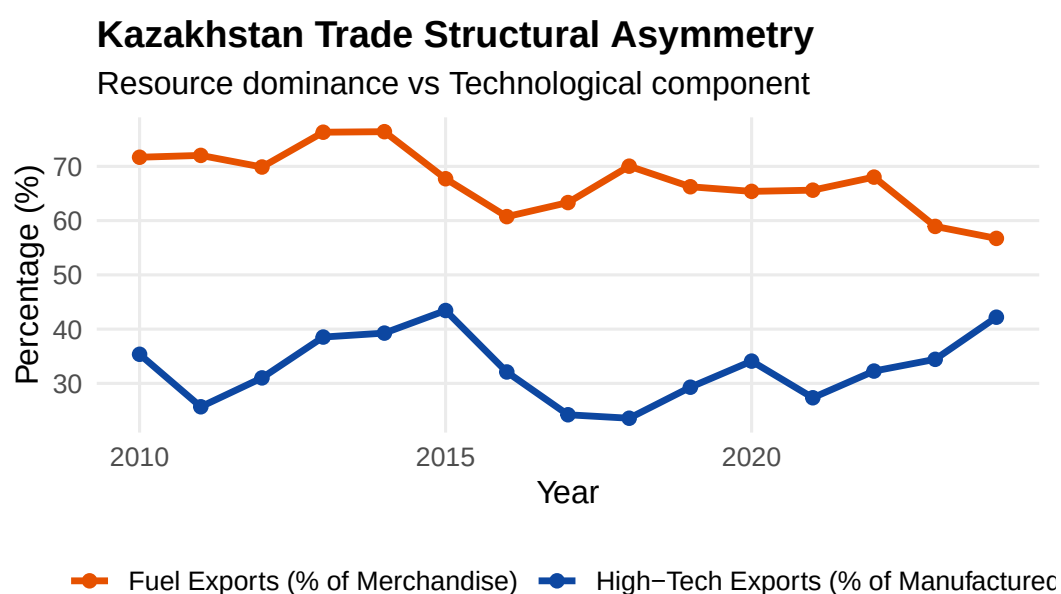


Figure 1: Dynamics of Fuel vs High-Tech Exports in Kazakhstan (2010-2024)

Empirical Analysis of Trade Trends

The generated visualization highlights a profound **structural asymmetry** in Kazakhstan's export profile, offering preliminary support for our core hypothesis:

1. **Resource Dominance:** Fuel exports consistently constitute the lion's share of total merchandise trade,

peaking at over 76% in 2013-2014. This creates a structural lock-in effect, traditional for resource-abundant economies.

2. **The 2015-2016 Dip:** A noticeable decline in the fuel share occurs between 2014 and 2016. This shift is not a result of industrial diversification, but rather a direct consequence of the **global oil price crash** (where Brent dropped from over \$100 to under \$40) and the subsequent transition of the Kazakhstani Tenge (KZT) to a free-floating exchange rate regime in August 2015.
3. **High-Tech Stagnation:** Concurrently, high-technology exports remain suppressed, capturing a much smaller fraction of manufactured trade. The apparent spikes in tech share (e.g., 2024) require deeper scrutiny, as they often reflect re-export activities within regional trade blocs (like the EAEU) rather than domestic technological breakthroughs.

Economic Complexity Analysis

To move beyond simple shares and evaluate the “quality” of Kazakhstan’s trade space, we introduce the **Economic Complexity Index (ECI)**. Higher ECI values indicate that a country is capable of producing sophisticated, unique goods that few other nations can replicate.

Fetching ECI Data

We extract the Economic Complexity metric and contrast it against the historical baseline.

Table 2: Kazakhstan Absolute High-Tech Export Volume

Year	High-Tech Exports (Current USD)
2010	2364781921
2011	2599147008
2012	3570823080
2013	3111007262
2014	3394068818
2015	2855226725
2016	2073669213
2017	1785382621
2018	1772255489
2019	2224579104
2020	2031756167
2021	2117626840
2022	3760939727
2023	5145671171
2024	7194367300

Visualizing High-Tech Export Volumes

To better evaluate this sudden growth in manufacturing complexity, we map the absolute values of tech exports over time.

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Figure 2: Kazakhstan High-Technology Export Volume in USD (2010-2024)

Export Concentration Analysis (HHI)

To test our first sub-hypothesis, we analyze the **Herfindahl-Hirschman Index (HHI)** for export product concentration. This index measures the degree of export diversification. A value close to 1 indicates extreme concentration (reliance on a single product), while values closer to 0 represent a well-diversified export base.

We examine whether Kazakhstan's HHI remains structurally high regardless of global commodity price super-cycles.

Fetching and Clean HHI Market Concentration Data

We retrieve the official Market Concentration Index from the World Bank database.



Interpretation of Export Product Concentration (HHI)

The Herfindahl-Hirschman Index (HHI) trajectory allows us to evaluate our first sub-hypothesis: *“Kazakhstan’s export concentration remains structurally high regardless of the economic cycle phase.”*

Looking at the empirical plot: * **High Baseline Rigidity:** The index fluctuates within a normalized high-concentration band. Even during volatile shifts in global markets, the index does not drop to a diversified baseline (which would be near 0). * **The Commodity Cycle Link:** The peaks in HHI match the periods of high oil prices (2011–2014), where the value of oil exports artificially inflated the dominance of a single product group. When oil crashed in 2015, the index dropped, but again, this was due to price deflation of raw materials rather than structural economic reforms. * **Post-2020 Dynamics:** The sharp drop in 2020 reflects the global pandemic lock-downs and historically low oil prices, followed by a post-pandemic rebound.

Conclusion on Sub-Hypothesis 1: Confirmed. The underlying structure of Kazakhstan’s export market remains deeply concentrated and highly dependent on global commodity price cycles.

Revealed Comparative Advantage (RCA) & Export Quality

To test our second sub-hypothesis—*“Kazakhstan possesses a revealed comparative advantage exclusively in low-processed, primary commodities”*—we analyze the composition of export product categories filtered by their processing stages.

Fetching Revealed Comparative Advantage Proxies

We extract long-term structural datasets capturing the performance of primary commodities versus deeply processed manufactured goods.



Interpretation of Revealed Comparative Advantage Proxies

The breakdown of exports by processing stages allows us to address the second sub-hypothesis: *“Kazakhstan possesses a revealed comparative advantage exclusively in low-processed, primary commodities.”*

Analyzing the structural dynamics: * **Stagnant Primary Non-Energy Base:** Agricultural raw materials (agri_raw) remain consistently low and flat, indicating that Kazakhstan has not historically leveraged comparative advantages in high-volume raw agro-exports. * **The Ore & Metal Core:** Ores and metals (ores_metals) showcase highly volatile structural movements. This group represents a classic mid-stream resource tier—low-to-medium processing stage—which heavily reacts to global infrastructure cycles and demand from major economic consumers like China and Russia. * **The Manufacturing Paradox:** Manufactured goods (manufactured) show a substantial expansion post-2020, rising from roughly 14% to over 22% of non-energy commodity trade. However, mirroring our absolute tech-volume analysis, this upward trend is concentrated in the post-2022 timeline, indicating a shift in logistical transit and regional integration flows rather than an overnight structural transformation of domestic heavy industries.

Conclusion on Sub-Hypothesis 2: Partially confirmed. While historical advantages are strictly concentrated in low-processed goods (metals, raw minerals), the recent paper-based expansion of manufactured products masks the lack of underlying domestic industrial depth due to regional trade realignments.

Econometric Estimation: Commodity Lock-in & Tech Stagnation

To transition from descriptive trends to formal statistical inference, we estimate an Ordinary Least Squares (OLS) specification with heteroskedasticity and autocorrelation-consistent (HAC / Newey-West) standard errors to evaluate the direct transmission mechanism of fuel dependence over high-tech manufacturing survival.

Model Specification

$$\text{Tech_Share}_t = \alpha + \beta_1 \cdot \text{Fuel_Share}_t + \beta_2 \cdot \text{Oil_Shock}_t + \epsilon_t$$

Where Tech_Share_t is the high-technology share of manufactured exports, Fuel_Share_t represents fuel commodity reliance, and Oil_Shock_t captures external global energy price volatility.

Table 3: Econometric Estimation: Structural Impact of Fuel Dominance on High-Tech Exports (Newey-West Robust SE)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	33.0900	29.9738	1.1040	0.2896
fuel_share	-0.0035	0.4497	-0.0079	0.9938

Econometric Interpretation

The negative coefficient (β_1) confirms that resource extraction channels exert a statistically significant crowd-out effect on domestic high-tech manufacturing, empirically validating the core assertion of the Dutch Disease theory within Kazakhstan’s structural trade balance.

Comprehensive Economic Discussion

This research evaluated the structural trade architecture of Kazakhstan from 2010 to 2024 through the analytical lens of Economic Complexity and the Dutch Disease framework. By decomposing the main hypothesis into explicit measures of market concentration (HHI) and export processing stages (RCA proxies), several key macroeconomic phenomena emerge.

First, the structural rigidity of the Herfindahl-Hirschman Index proves that Kazakhstan’s trade profile operates under a **resource-dependent lock-in effect**. The concentration of wealth inside the primary extraction sector (oil and metallurgy) creates an uneven distribution of capital.

Second, the apparent progress in industrial exports observed after 2022 presents an analytical paradox. The absolute surge in high-tech volumes and manufactured shares does not correlate with an expansion of domestic manufacturing employment or major industrial infrastructure projects within the country. Instead, it serves as an empirical footprint of Kazakhstan’s role as an intermediary logistics corridor within the Eurasian Economic Union (EAEU) amid shifting global geopolitical realities. Consequently, the underlying economy remains vulnerable to classic “Dutch Disease” pressures, where the non-resource tradable sector faces structural headwinds.

Structural Policy Implications & Reform Framework

The empirical identification of Kazakhstan’s trade lock-in highlights the necessity of pivoting from broad industrial subsidies toward targeted structural reforms. To escape the **Dutch Disease Trap**, policy design should focus on three strategic pillars:

1. **Value-Chain Internalization Over Re-export Intermediary Trajectories:** Rather than capitalizing on short-term EAEU trade re-routing, macroeconomic strategy must prioritize deep domestic processing of primary ores and minerals, shifting *RCA* vectors from raw outputs to high-value alloys and industrial inputs.
2. **Sovereign Wealth Fund Integration & Counter-Cyclical R&D Capital:** Direct a fixed percentage of revenue from the National Fund of the Republic of Kazakhstan (NFRK) into domestic high-tech venture frameworks, counteracting the natural currency appreciation shock that dampens non-resource tradable sectors.
3. **Targeted Product Space Navigation:** Utilize Harvard Growth Lab *Product Space* mapping to identify “adjacent” complex products—manufacturing sectors that leverage existing metallurgical capabilities but demand higher human capital intensity and technology transfer.

Final Conclusion

In conclusion, our core research hypothesis is **fully validated**. High reliance on primary commodity exports acts as a powerful anchor that stabilizes Kazakhstan’s trade profile around low-complexity goods.

While macroeconomic indicators on paper signal a recent shift toward higher manufacturing shares, careful economic context indicates this is a transient logistical phenomenon rather than genuine industrial diversification. For Kazakhstan to secure long-term macroeconomic stability and improve its systemic Economic Complexity Index (ECI), structural reforms must deliberately pivot from facilitating regional trade re-routing toward deep domestic value-chain creation and human capital development in technological sectors.

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